

Gravity

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

	QUESTION	ANSWER	✓
1	Define weight.	The force acting on an object due to gravity.	☐
2	State why objects experience weight on Earth.	The Earth's gravitational field attracts them.	☐
3	Explain why objects experience different weights on different planets.	Each planet has a different gravitational field strength.	☐
4	Does the mass of an object change if it is moved to another planet?	No	☐
5	Does the weight of an object change if it is moved to another planet?	Yes: the gravitational field strength is different.	☐
6	State the standard unit of weight.	Newton (N)	☐
7	State the standard unit of mass.	Kilogram (kg)	☐
8	State the standard unit of gravitational field strength.	Newton per kilogram (N/kg)	☐
9	State the mathematical relationship between weight and mass.	Weight is directly proportional to mass ($W \propto m$).	☐
10	State the meaning of this symbol: $\propto$	Is directly proportional to	☐
11	Describe how the weight of an object depends on its mass.	Doubling the mass doubles the weight (direct proportion).	☐
12	What is the centre of mass of an object?	The single point where the weight of the object can be considered to act.	☐
13	State the equation linking weight, mass and gravitational field strength.	weight = mass $\times$ gravitational field strength ($W = m g$)	☐
14	Name the instrument used to measure weight.	A newtonmeter (calibrated spring balance)	☐

QUESTION		ANSWER	✓
15	State the gravitational field strength at the surface of the Earth.	About 9.8 N/kg	☐